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Paper 1: A semi-mechanistic population pharmacokinetic-pharmacodynamic model to assess do

Bibliographic Information

- **Title:** A semi-mechanistic population pharmacokinetic-pharmacodynamic model to assess downstream drug-target effects on erythropoiesis
- **Authors:** Rognås SV, Schaedeli Stark F, Marchesi M, et al.
- **Journal:** Journal of Pharmacokinetics and Pharmacodynamics, 2025, 52:42
- **DOI:** <https://doi.org/10.1007/s10928-025-09990-7>

Paper Type: POPK_PKPD

Quick Take

This paper presents a semi-mechanistic PKPD model of erythropoiesis and hemoglobin synthesis based on bitopertin data. It is essential reading for pharmacometricians working on hematological drug effects, as it provides a versatile framework for simulating various drug-target interactions on red blood cell dynamics. The model can inform dose selection and safety assessments in drug development.

Executive Summary

Rognås et al. developed a population PKPD model integrating reticulocyte, erythrocyte, MCH, and IRF data from a 120-day bitopertin study in healthy subjects. The model uses transit compartments to capture maturation stages and hemoglobin synthesis, with feedback mechanisms mimicking erythropoietin response. It accurately described dose- and time-dependent changes and was used to simulate four hypothetical mechanisms of drug action, demonstrating its utility for predicting anemia risk and guiding clinical development. The work bridges data-driven modeling with physiological knowledge, offering a reusable framework for assessing hematological safety.

Scientific Context & Motivation

Erythropoiesis is tightly regulated, and drug-induced disruptions can lead to anemia. Existing models often focus on hemoglobin turnover alone, lacking integration of short-term biomarkers like reticulocytes and IRF. This study fills the gap by combining multiple hematological endpoints into a single semi-mechanistic model, enabling prediction of downstream effects on hemoglobin and erythrocyte counts from early signals. It also addresses the need for a flexible platform to test hypothetical drug-target interactions across different stages of erythropoiesis.

✂ Methodological Snapshot

The model is a semi-mechanistic population PKPD model using transit compartments to represent erythropoiesis stages: precursor, reticulocyte maturation in bone marrow and blood, and

erythrocyte aging. Hemoglobin synthesis is modeled as a parallel chain. Drug effect (bitopertin) inhibits hemoglobin synthesis rate. Homeostatic feedback on precursor recruitment and reticulocyte release is driven by Hb_{tot} deviation. An empirical tolerance mechanism accounts for stem cell depletion. Model fitted simultaneously to RET, RBC, MCH, IRF using NONMEM with SAEM estimation.

☒ Structural Model Breakdown

The model comprises: (1) a precursor compartment in bone marrow with transit time $LSPRE = 1/k_{PRE}$; (2) a four-compartment reticulocyte maturation chain split between bone marrow and blood, parameterized by k_{RET} and the fraction of premature release (*prelease*, linked to IRF); (3) a four-compartment erythrocyte aging chain with transit rate $k_{RBC} = n_{CTR}/LS_{RBC}$; (4) a parallel four-compartment chain for mean corpuscular hemoglobin (MCH) in each erythrocyte compartment, with hemoglobin synthesis rate $R_{in,MCH}$ inhibited by bitopertin via an I_{max} function on AUC_{ss} ; (5) homeostatic feedback: an exponential function of fractional Hb_{tot} change stimulates precursor recruitment ($R_{in,PRE}$) and k_{RET} ; (6) an empirical tolerance mechanism using two compartments that modulates $R_{in,PRE}$ based on precursor accumulation. Total hemoglobin is the sum of $RBC_n \times MCH_n$ across compartments. The model simultaneously fits RET, RBC, MCH, and IRF observations.

Detailed Methodological Analysis

Modeling Approach

A semi-mechanistic transit compartment model was developed, expanding a previous hemoglobin turnover model. It includes a precursor compartment with transit time $LSPRE$, a four-compartment reticulocyte maturation structure (bone marrow and blood), and a four-compartment erythrocyte aging chain. Hemoglobin content is tracked in parallel transit compartments. Drug effect is an exposure-dependent inhibition of hemoglobin synthesis rate (I_{max} model). Homeostatic feedback stimulates precursor recruitment and reticulocyte release as a function of Hb_{tot} deviation. An empirical tolerance mechanism modulates precursor recruitment based on relative precursor amount. The model was implemented in NONMEM 7.4.3 with PsN and mrgsolve for simulations.

Data Sources

Data from a Phase 1, randomized, double-blind, placebo-controlled trial in 67 healthy subjects (aged <50 years) receiving bitopertin 10, 30, or 60 mg or placebo once daily for 120 days, with a 120-day follow-up. Hematological assessments included reticulocyte count (RET), immature reticulocyte fraction (IRF), erythrocyte count (RBC), mean corpuscular hemoglobin (MCH), and total hemoglobin (Hb_{tot}). Rich PK sampling allowed estimation of individual AUC_{ss} as the exposure metric. 62 subjects were included in the final analysis.

Estimation Methods

First-order conditional estimation with interaction (FOCE-I) was used initially; as model complexity increased, Stochastic Approximation Expectation-Maximization (SAEM) was employed with 3000 burn-in and 1000 accumulation iterations. Importance Sampling (IMP) evaluated the objective function. Mu-referencing was used for efficiency. Standard errors were obtained from the observed Fisher information matrix.

Model Evaluation

Model selection used OFV ($p < 0.01$ for hierarchical models) and AIC for non-hierarchical comparisons. Parameter precision was assessed via RSE (<40% fixed, <50% random). Condition number < 1000 indicated no overparameterization. Goodness-of-fit plots, CWRES, and visual predictive checks (VPCs, 200 simulations, 95% prediction and confidence intervals) evaluated predictive performance. Sensitivity analysis was performed for outliers ($CWRES > 5$).

Covariate Analysis

No formal covariate analysis was performed beyond dose as a structural component. The model incorporated dose-dependent drug effect via AUC_{ss} . The study population was homogeneous (healthy, <50 years), limiting exploration of covariates like age, sex, or weight.

Statistical Rigor Assessment

The statistical approach is appropriate for the complex PKPD model. The use of SAEM with importance sampling and mu-referencing is state-of-the-art for high-dimensional nonlinear mixed-effects models. Model evaluation included standard diagnostics (GOF, VPCs) and sensitivity to outliers. The sample size (62 subjects) is adequate for estimating population parameters, though limited for covariate exploration. Missing data were handled by the likelihood-based estimation. The empirical tolerance component, while improving fit, adds parameters that may be data-driven; its identifiability was not extensively discussed. The lack of external validation is a limitation, but internal predictive checks (including prediction of unused Hb_{tot} data) provide some confidence.

Key Findings

The model estimated an erythrocyte lifespan of 125 days, consistent with literature. Bitopertin inhibited hemoglobin synthesis, causing a dose-dependent decrease in MCH and a compensatory increase in reticulocyte production via feedback. The model captured the delayed erythrocyte response and predicted Hb_{tot} changes not used in fitting. Simulations of four mechanisms revealed distinct patterns: inhibition of hemoglobin synthesis (Mechanism A) led to continuous Hb_{tot} decline; inhibition of precursor production (B) caused a new steady state; inhibition of maturation (C) showed gradual decline; and combined inhibition with feedback suppression (D) resulted in Hb_{tot} below anemia threshold in some subjects. These findings highlight the importance of mechanism-specific modeling for safety assessment.

🔗 Clinical & Regulatory Implications

The model can predict the time course of hemoglobin and erythrocyte counts under various drug mechanisms, aiding in dose selection to avoid anemia. For bitopertin, the continuous decline in Hb_{tot} under Mechanism A suggests that long-term treatment may require monitoring beyond 120 days. The simulations indicate that inhibition of precursor production (Mechanism B) leads to a new steady state, whereas combined inhibition of hemoglobin synthesis and feedback (Mechanism D) poses the highest anemia risk. The framework can be adapted to other compounds, supporting regulatory submissions by providing mechanistic evidence of hematological safety. It also enables hypothesis testing for early clinical development, potentially reducing the need for long-term trials.

Strengths & Limitations

Strengths

- Rich longitudinal dataset covering full erythrocyte lifespan
- Integration of multiple biomarkers (RET, IRF, RBC, MCH) into a single model
- Physiologically plausible structure with feedback and tolerance mechanisms
- Simulation of hypothetical mechanisms demonstrates practical utility
- Model predicted Hb_{tot} well without using it for fitting

Limitations (Acknowledged by Authors)

- No external validation with other compounds
- Age restriction to <50 years limits generalizability to older populations
- Empirical nature of feedback and tolerance components may affect extrapolation
- Iron metabolism not explicitly modeled

Limitations (Expert Review)

- The model assumes equal maturation rates in bone marrow and blood, which may oversimplify
- The tolerance mechanism is empirical and may not fully capture stem cell dynamics
- No covariate analysis beyond dose; potential effects of sex, weight, etc. not explored
- The model was developed on healthy subjects; applicability to disease states (e.g., anemia of chronic disease) is uncertain

Generalizability

The model is generalizable to other compounds affecting erythropoiesis, provided the mechanism aligns with the modeled pathways. However, extrapolation to patient populations with altered erythropoiesis (e.g., renal impairment, inflammation) requires careful validation and potential extension with iron metabolism or disease-specific modules.

Key Equations

Precursor lifespan

$$LS_{PRE} = \frac{1}{k_{PRE}}$$

Transit time of precursor cells in the bone marrow, where k_{PRE} is the first-order rate constant for precursor maturation.

Reticulocyte maturation rate constant

$$k_{RET} = \frac{k_{RBC} \left(\frac{RBC_0}{n_{CTR}} \right)}{RET_0(1 - IRF_0)}$$

Derivation of the reticulocyte maturation and migration rate constant from baseline erythrocyte count (RBC_0), reticulocyte count (RET_0), immature reticulocyte fraction (IRF_0), and erythrocyte transit rate (k_{RBC}).

Erythrocyte transit rate constant

$$k_{RBC} = \frac{n_{CTR}}{LS_{RBC}}$$

First-order rate constant for erythrocyte aging through four transit compartments, where $n_{CTR} = 4$ and LS_{RBC} is the erythrocyte lifespan.

Total hemoglobin

$$Hb_{tot} = \sum_{n=1}^4 RBC_n \times MCH_n$$

Total hemoglobin concentration calculated as the sum of products of erythrocyte count and mean corpuscular hemoglobin in each of the four aging compartments.

Figures & Tables

- **Figure 1:** Longitudinal profiles of reticulocyte count, IRF, erythrocyte count, Hb_{tot} , and MCH for placebo and bitopertin dose groups.
 - *Significance:* Illustrates the dose- and time-dependent changes in hematological biomarkers, providing the basis for model development.
- **Figure 2:** Schematic of the semi-mechanistic model structure, including precursor, reticulocyte maturation, erythrocyte aging, and hemoglobin synthesis compartments.
 - *Significance:* Visualizes the model's physiological compartments and feedback loops, essential for understanding the modeling framework.
- **Figure 3:** Visual predictive checks (VPCs) for reticulocyte count, IRF, erythrocyte count, and MCH stratified by dose.
 - *Significance:* Demonstrates the model's predictive performance and its ability to capture central trends and variability.
- **Figure 4:** Comparison of model-predicted versus observed total hemoglobin (Hb_{tot}) over time, though Hb_{tot} was not used in model fitting.

- *Significance*: Validates the model’s ability to predict a clinically relevant endpoint not directly included in the estimation, supporting its utility for anemia risk assessment.
 - **Table 1**: Summary of demographics and baseline hematological biomarkers for the study population.
 - *Significance*: Provides context on the healthy subject population and the data available for model development.
 - **Table 2**: Parameter estimates, inter-individual variability, and residual error for the final PKPD model.
 - *Significance*: Reports the quantitative model parameters with precision, enabling reproduction and application of the model.
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Supplementary Materials

Supplementary material (Supplementary A1) provides additional details on the simulated drug-target interaction mechanisms (A–D), including equations and parameter settings used for the hypothetical simulations.

Future Directions

Future work should validate the model with external data from compounds with different mechanisms (e.g., iron chelators, EPO mimetics) and in patient populations. Incorporating iron metabolism and reticulocyte hemoglobin content (CHr) could enhance predictive power. Extending the model to account for blood loss, transfusions, and disease states would broaden its clinical applicability. Additionally, exploring covariate effects (age, sex, renal function) in larger datasets would improve individualized predictions.

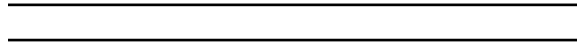
Expert Commentary

This work exemplifies the power of semi-mechanistic modeling in translating early biomarker signals into clinically meaningful endpoints. The use of a rich Phase 1 dataset with long-term dosing is rare and provides a unique opportunity to characterize erythropoietic dynamics. The model’s ability to simulate hypothetical mechanisms is a clever way to demonstrate its versatility, but the empirical feedback and tolerance components warrant caution when extrapolating beyond the studied dose range. The lack of external validation is a notable gap, but the physiological consistency of parameter estimates lends credibility. Overall, this is a well-executed study that advances the field of hematological PKPD modeling and sets a standard for integrating multiple biomarkers.

Bottom Line

This semi-mechanistic PKPD model provides a robust framework for predicting hematological effects of drugs that interfere with erythropoiesis. By integrating reticulocyte dynamics and hemoglobin synthesis, it enables early detection of anemia risk and supports dose optimization. Pharmacometricians can adapt this model to new compounds by mapping their mechanism of

action to the appropriate pathway, making it a valuable tool for model-informed drug development in hematology.



☒ Figures

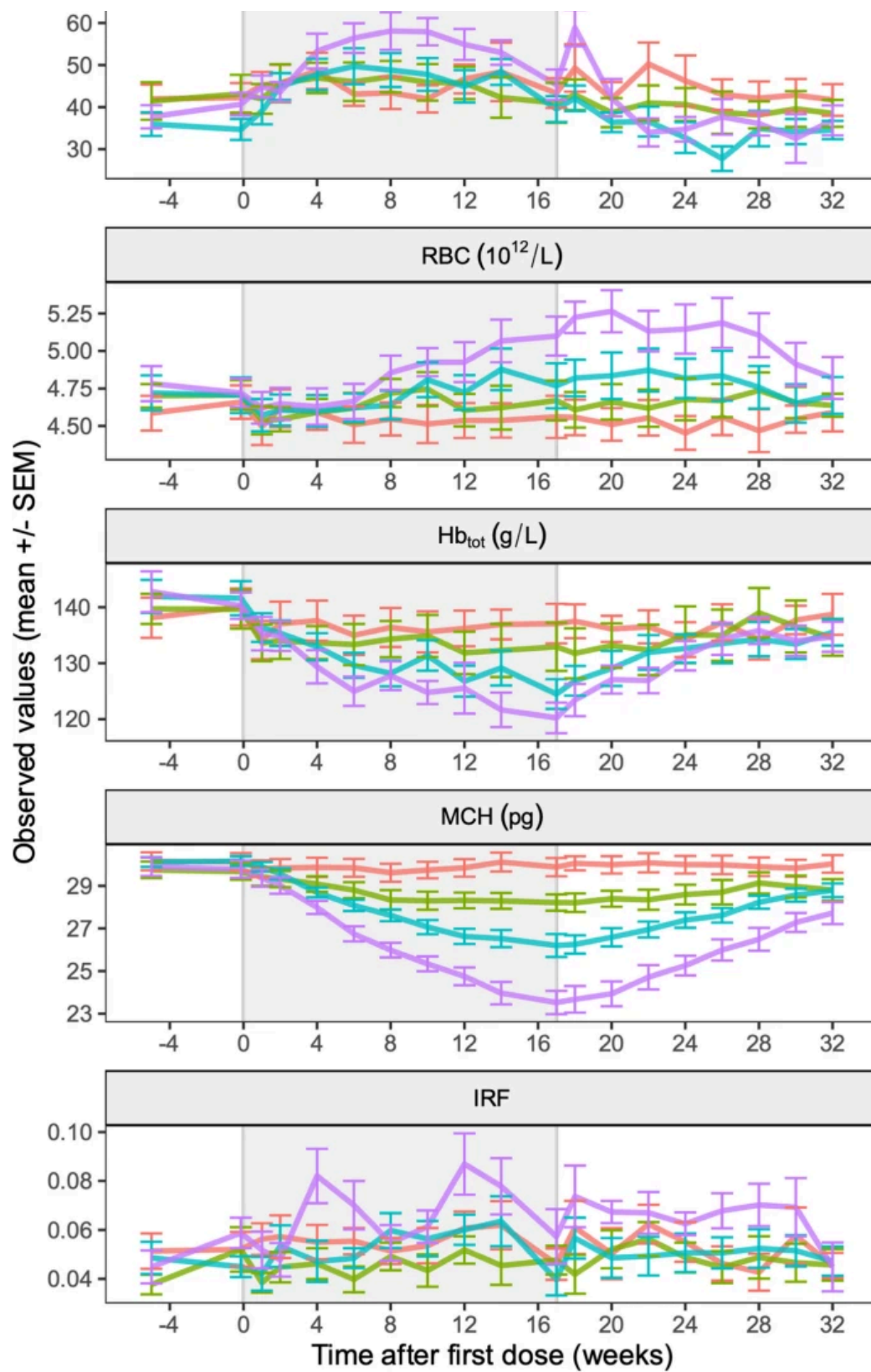


Figure 1: Figure 1

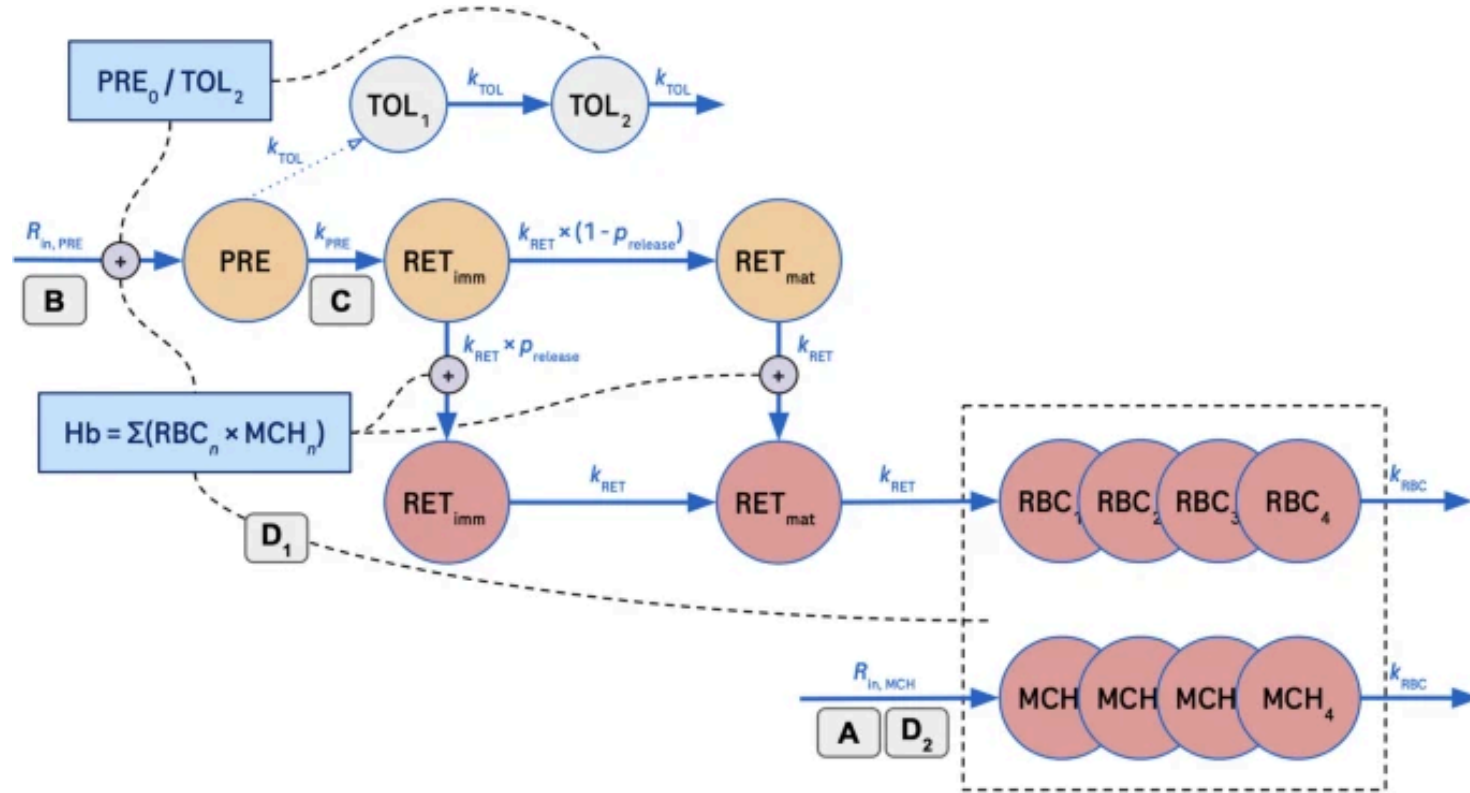


Figure 2: Figure 2

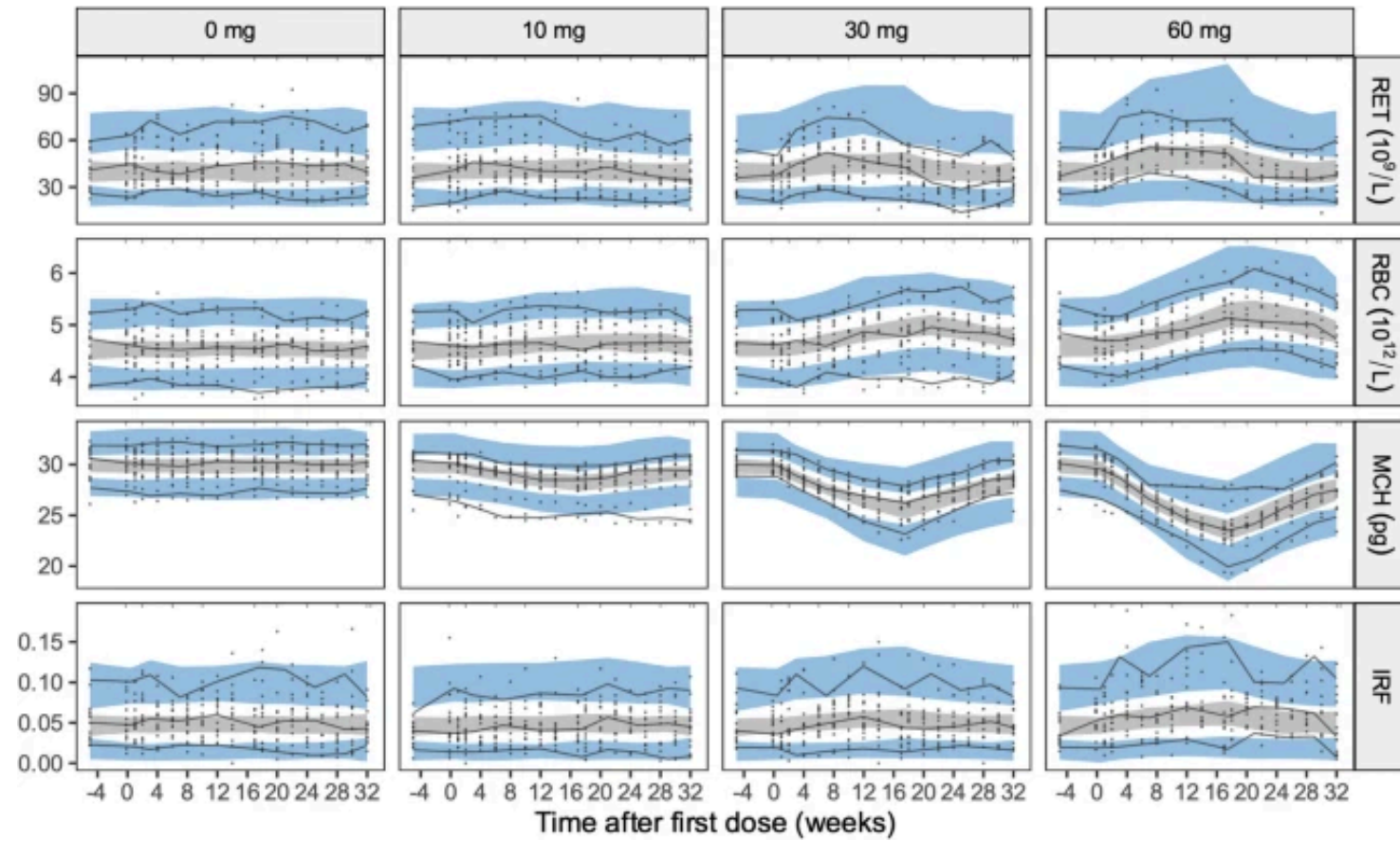


Figure 3: Figure 3

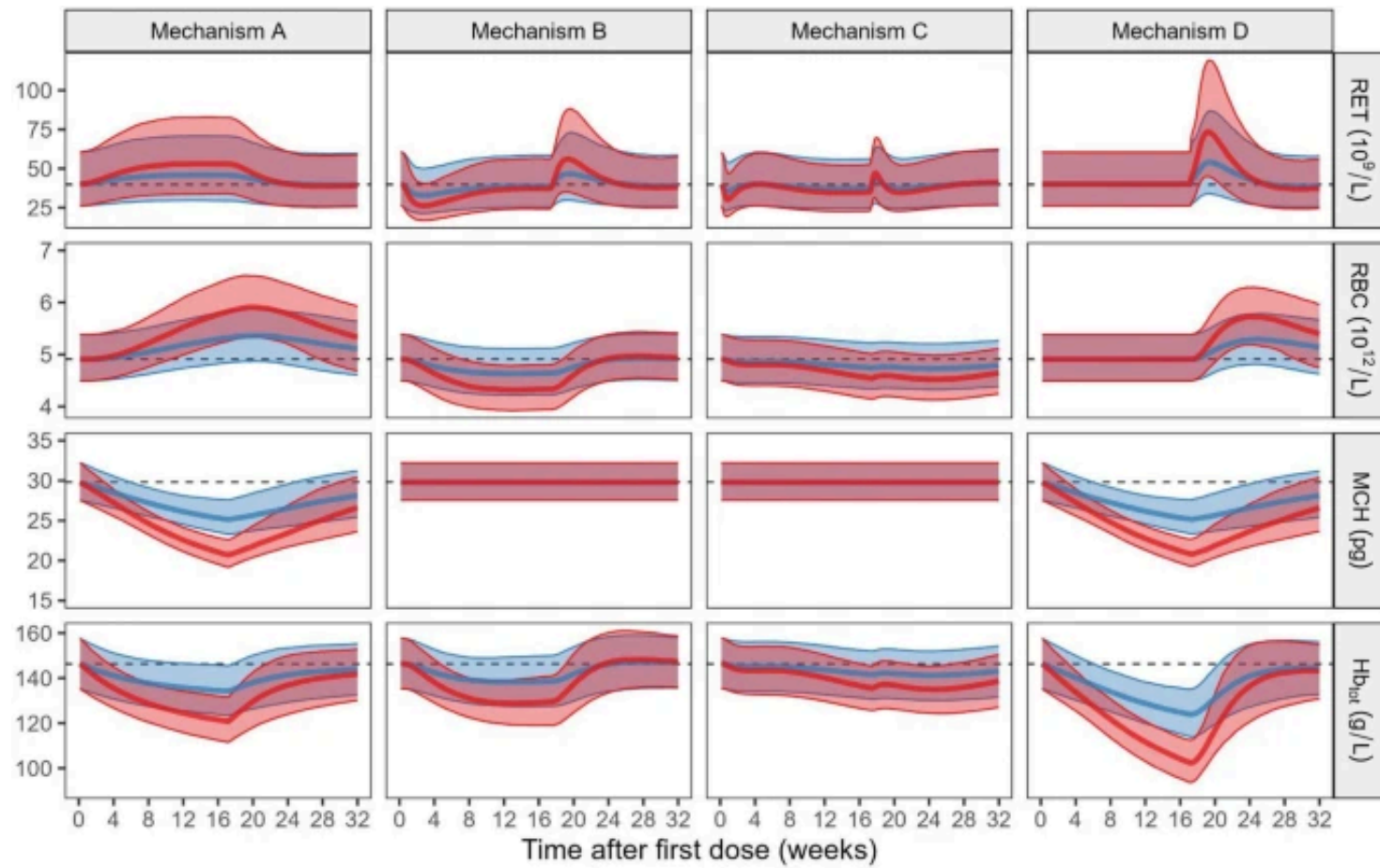


Figure 4: Figure 4